# Jake Golden

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## Education

**Duke University — Pratt School of Engineering** | Durham, NC
*Aug 2023 - Present*

- B.S.E. in Biomedical Engineering & Electrical and Computer Engineering, expected May 2027
- Cumulative GPA: 4.00/4.00

## Experience

**Analog Design Intern** | National Instruments (Emerson), Austin, TX
*May 2026 - Aug 2026*

- Designed schematic and layout for precision LCR/SMU characterization PCB.
- Implemented low-leakage switching architecture in guarded multiplexed signal chain.
- Authored 15+ V&V tests for LCR/SMU meter, automated instrumentation control and data acquisition/processing using LabVIEW, compiled results in report.

**Product Development Intern** | Stryker, Mahwah, NJ
*May 2025 - Aug 2025*

- Iteratively developed 9 prototypes for novel knee replacement implant: created design inputs to address unmet user needs and modeled in PTC Creo.
- Verified geometric tolerancing and dimensioning equivalency to legacy devices.
- Automated surgical template generation (MATLAB/Creo), saving 10+ hours of work.

**Computational Neuroscience Researcher** | Brain Stimulation Engineering Lab at Duke University Medical Center, Durham, NC
*Jan 2024 - Present*

- Program finite element analysis (FEA) ECT/TMS brain stimulation models in MATLAB
- Develop automated pipelines (MATLAB/Bash) that integrate MRI processing, mesh generation/processing (4M+ elements), and coil optimization for high-throughput, patient-specific FEA simulations (~50 simulations/week).

**Mechanical Engineering Intern** | SIMS Pump Valve Company, Hoboken, NJ
*Sep 2022 - Jun 2023*

- Created ISO-standard CAD models/drawings using surface modeling in SOLIDWORKS.
- Built standardized part library (300+ parts) for efficient modeling in future projects.
- Performed tolerance verification on pump parts using calipers and micrometers.

## Personal Projects | *Visit https://jake-golden.github.io for more*

**MATLAB Pulse Oximeter**

- Designed single-supply AFE converting photodiode current to PPG waveform: TIA + 0.8–5 Hz conditioning chain + biased gain sized to fill ADC's dynamic range
- Implemented custom bidirectional Arduino-MATLAB serial protocol with fixed-length packets, start-byte framing, and checksum validation
- Developed MATLAB DSP (filtering, amplitude extraction, R-ratio-to-$SpO_2$ calibration) and GUI layer with live plotting, LED control, and diagnostics console

**PID-Controlled Self-Balancing Robot**

- Built two-wheeled self-balancing robot using Arduino, IMU sensors, and DC motors
- Implemented $I^2C$ communication with custom PCB and housing.
- Iteratively tuned PID control parameters to achieve stable balancing.

## Skills

**Analog/Digital Circuit Design**

- SPICE simulation (LTSpice)
- Op-amp circuit design, filter design, feedback, stability analysis
- PCB Design: Schematic capture and layout (KiCad, Xpedition)
- Basic CPU design, combinational/sequential Boolean logic
- Power electronics & power supply design
- *In progress*: Analog IC design and layout (Cadence Virtuoso)

**Mechanical Design**

- CAD: SOLIDWORKS, PTC Creo, Autodesk Fusion
- Fabrication: 3D printing (FDM), laser cutting

**Programming**

- MATLAB (simulation, automation, FEA)
- Python, C, Java, LabVIEW

**Applied Mathematics**

- **Signal processing**: Time/frequency-domain analysis, filters, control systems
- **Wave systems**: Fourier analysis, transmission line theory, electromagnetic waves
- **Experimental design**: Hypothesis testing, inferential statistics

## Awards & Honors

Pratt School of Engineering Dean's List with Distinction (2024, 2025, 2026)

## Activities & Interests

Duke University football team equipment manager

Part-time math tutor

Piano, guitar

Photography

College basketball enthusiast